A New Ceramic Magnetic Regenerator Material for 4 K Cryocoolers

T. Numazawa[1], T. Yanagitani[2], H. Nozawa[2], Y. Ikeya[3], R. Li[3] and T. Satoh[3]

[1]Tsukuba Magnet Laboratory, National Institute for Materials Science
3-13 Sakura, Tsukuba 305-0003, Japan
[2]Ceramics Division, Konoshima Chemical Co. Ltd.
80 Koda, Takuma-Cho, Mitoyo-Gun, Kagawa 769-1103, Japan
[3]Cryogenics Department, Sumitomo Heavy Industries, Ltd.
2-1-1 Yato-Cho, Nishitokyo City, Tokyo 188-8585, Japan

ABSTRACT

A new class of magnetic regenerator materials for 4 K cryocoolers has been developed. A hexagonal gadolinium oxisulfide Gd_2O_2S = GOS has been found to provide a very high volumetric heat capacity of more than 1 J/cm³K at 5.2 K. Polycrystal ceramic GOS particles have been fabricated with diameters between 0.35 mm and 0.45 mm. The achieved smooth surface on the spherical particles and a Vickers hardness of ~900 make the material very suitable for use in regenerators.

The measured data on the thermal and magnetic properties of the GOS also show excellent characteristics as a regenerator material; no change in heat capacity is observed when applying a magnetic field up to 1 T, and the high thermal conductivity of ~0.05 W/cmK at 6 K is twice that of GAP. Magnetization of GOS at 1 T is about half that of $HoCu_2$ between 2 K and 20 K.

System-level cooling tests have been conducted with both GM-cycle and pulse tube cryocoolers where a portion of the $HoCu_2$ in the 2nd regenerator was replaced with GOS. At a 50% volumetric ratio of GOS to $HoCu_2$, the cooling power increased up to ~25 % at 4.2 K. The cooling power did not drop above 4.5 K, where it significantly decreases when GAP is used. The new ceramic regenerator material GOS is expected to not only contribute to increased cooling performance at 4.2 K, but also to decreased fabrication cost.

INTRODUCTION

Current technology development on conventional mechanical cryocoolers continues to achieve breakthroughs in performance and reliability. However, from the view point of materials research, new regenerator materials are still a key component to improving both the performance and the cost of cryocoolers, especially in the low temperature region below 4 K. We have investigated promising candidate materials focusing on rare-earth oxides, and a ceramic magnetic regenerator material GAP ($GdAlO_3$) has been developed for 2 K to 4 K cryocoolers.[1] Cooling tests using GAP have shown excellent improvement in the cooling performance below 4 K[2-4], but the cooling capacity is still somewhat depressed because of GAP's steep decline in heat capacity above 4 K. Therefore, a new magnetic regenerator material, similar to the GAP, is desired for 4 K cryocoolers to provide higher heat capacity in the region immediately above 4 K.

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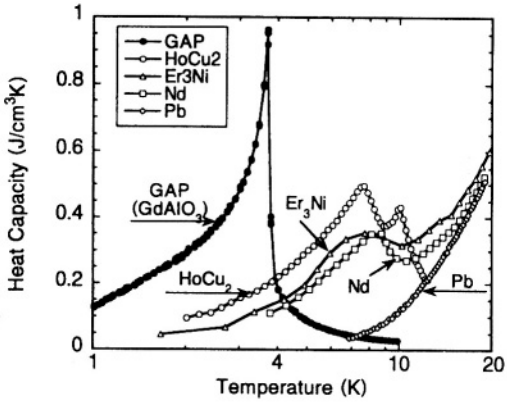


Figure 1. Heat capacity of GAP and other conventional regenerator materials.

This paper describes our approach to solving the requirement for a new 4 K regenerator material. Thermal and magnetic properties of the new ceramic magnetic material will be shown including its fabrication process. Then, experimental results of cooling tests with both GM and pulse tube cryocoolers will be presented to confirm the performance of the new magnetic material.

CERAMIC MAGNETIC REGENERATOR MATERIAL

Properties of Rare-Earth Oxides Used in Regenerators

Using oxide magnetic materials for low temperature regenerators was first suggested in our previous papers.[1,5] The developed GAP material, shown in Fig.1, displays a considerably higher heat capacity than other conventional metallic regenerator materials in the region below 4 K. Such a high heat capacity has proven to be very effective in increasing the cooling performance below 4 K; however, the heat capacity suddenly drops above 4 K because of the antiferromagnet properties of the material.

Several important characteristics of rare-earth oxides for regenerator applications are summarized below:

1) Rare-earth oxides tend to show antiferromagnetic behavior
2) Generally, the magnetic transition temperature is not high, roughly <10 K
3) Heat capacity shows very sharp lambda type anomaly
4) Magnetic field dependence of heat capacity is small for < 1 T fields
5) Magnetization is weak for < 1 T fields

It is noticeable that these properties show both positive and negative aspects as a regenerator material. For example, a sharp peak of the heat capacity is not desirable for a regenerator, but this comes from the anti-parallel alignment of magnetic moments in the oxide crystal structures. On the other hand, this same magnetic interaction provides excellent magnetic characteristics as described in 4) and 5) compared with other metallic regenerator materials.

Approach of Developing a New Ceramic Regenerator Material for 4 K

Figure 2 shows the results of a cooling test with a GM cryocooler where 20 % of the volume of $HoCu_2$ was replaced with GAP. It is clear that the GAP material increases the cooling power below 4 K, but there is minimal advantage to using the material above 4.2 K. Then, some questions arise as to what kind of heat capacity properties are most important for a 4 K regenerator. It can be pointed out that the peak temperature and the broadness of the heat capacity in a certain temperature

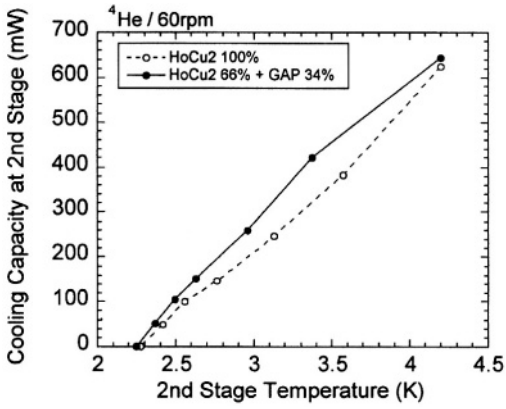


Figure 2. Experimental result on cooling capacity at 2nd regenerator with GM cryocooler by replacing 20 % volume of $HoCu_2$ with GAP.

range must be considered, but there is no clear answer except real cooling tests with cryocoolers.

However, numerical studies will contribute to predict the fundamental aspects of the heat capacity. Torii et. al. showed very useful simulation results on the heat capacity of regenerator material relating to the cooling capacity for 4 K cryocoolers.[6] From the view of oxide regenerator material, we summarize their calculated results as shown below:

1) Heat capacity must be larger than $0.3\ J/cm^3K$ to provide a cooling capacity of ~1 W.
2) Peak temperature of heat capacity between 4.2 K and 5 K is effective in increasing the cooling capacity at 4.2 K.
3) Temperature width of heat capacity peak is optimized between 2 K and 4 K to improve the cooling capacity at 4.2 K.

Increasing the peak temperature of GAP from 3.8 K to the temperature between 4.2 K and 5 K seems to be a most simple solution to provide better cooling performance. Thus, we first tried to modify the GAP by changing its composition. Figure 3 shows the heat capacity of 0.5 mol Gd-rich GAP, $Gd_{1.5}Al_{0.5}O_3$ in comparison with that of GAP. Twin peaks appear in $Gd_{1.5}Al_{0.5}O_3$ and this suggests that two crystals of Gd_2O_3 and $GdAlO_3$ coexist in the material. Solubility limit in

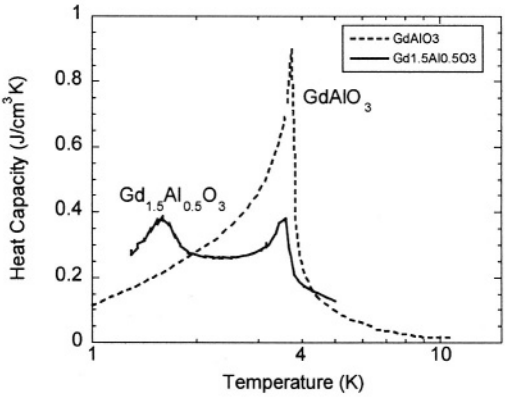


Figure 3. Heat capacity of 0.5 mol Gd-rich GAP, i.e., $Gd_{1.5}Al_{0.5}O_3$ and GAP.

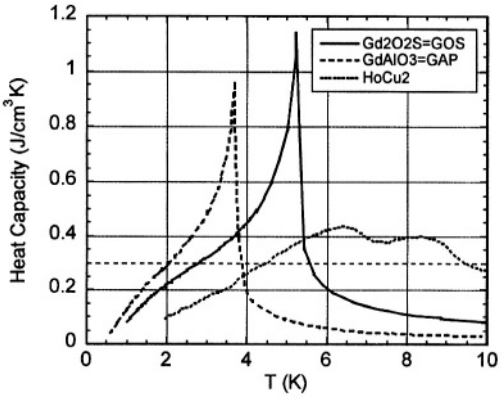


Figure 4. Volumetric heat capacity of GOS, GAP and $HoCu_2$.

the perovskite structure like GAP is very small and therefore, it is difficult to control the magnetic transition temperature by substituting Gd for the other rare-earth elements.

We investigated the alternate way to shift the peak temperature higher than 4.2 K based on the thermodynamic stability of the crystal growing process. Starting from a seed material Gd_2O_3, we identified that a hexagonal gadolinium oxisulfide Gd_2O_2S = GOS could be the best candidate for 4 K regenerator material because of the magnetic transition temperature of ~5 K and the high volumetric rare-earth density of the crystal structure.

Figure 4 shows the measured volumetric heat capacity of the GOS in comparison with the GAP and $HoCu_2$. The GOS has a considerably large heat capacity of more than 1 J/cm^{3}K at 5.2 K and the temperature width of the heat capacity peak is 2.5 K at 0.3 J/cm^{3}K, while the GAP has that of 1.8 K. Thus, it is concluded that the GOS satisfies the dominant conditions of the heat capacity as a regenerator material to improve the cooling performance of 4 K cryocoolers.

Thermal and Magnetic Properties of GOS

Figure 5 shows the magnetic field dependence of the heat capacity of the GOS. Zero magnetic field heat capacity shows a typical lambda type anomaly due to the antiferromagnetic material. There is no change in the heat capacity by applying the magnetic field up to 1 T. This

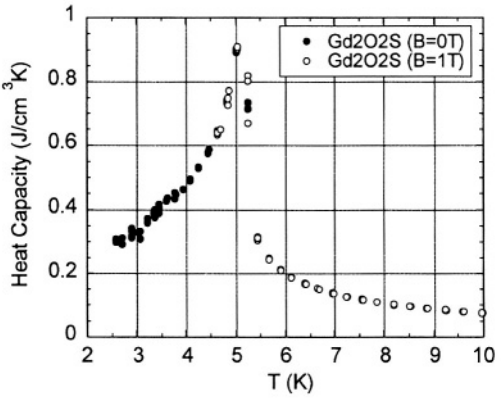


Figure 5. Magnetic field dependence of heat capacity of GOS for magnetic fields of 0 and 1 Tesla.

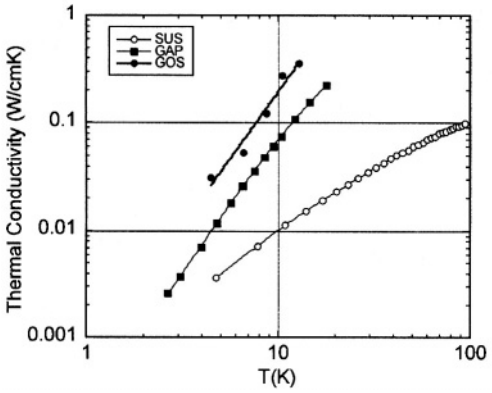


Figure 6. Thermal conductivity of ceramic GOS, ceramic GAP and SUS.

property is important for the 4 K cryocooler which is used to cool superconducting magnets.

Figure 6 shows the thermal conductivity data of the GOS at zero magnetic field in comparison with those of the GAP and a SUS. Note that the measured GAP and GOS samples are the polycrystal ceramics. The thermal conductivity of the GOS is higher than those of the GAP and the SUS, and it will contribute to shorten the thermal penetration time.

Figure 7 shows the magnetization data of the GAP as a function of temperature for the magnetic field of 1 T in comparison with that of $HoCu_2$. Magnetization of the GOS at 1 T is about twice smaller than that of $HoCu_2$ between 2 K and 20 K, Some 4 K cryocoolers are also used to cool the sensors which are sensitive to the magnetic noise like a SQUID, so the smaller magnetization of the regenerator material can reduce the magnetic noise coming from the cryocooler.

Fabrication of Ceramic Form GOS

In order to use GOS as the regenerator material, sphere form particles between 0.2 mm and 0.5 mm in diameter must be fabricated. The hardness of the particles is another important factor in avoiding the broken particles occasionally caused by the pressurized helium gas flow in the regenerator.

Basic scheme of the fabrication method of the ceramic GOS is very similar to that of GAP

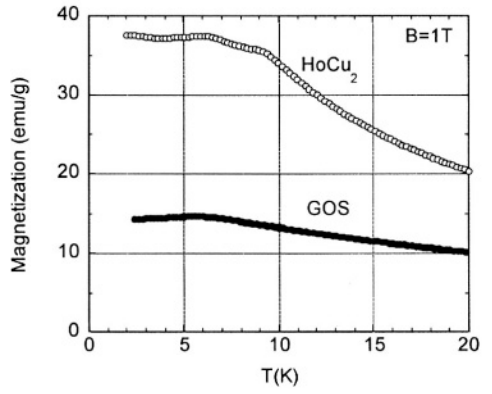


Figure 7. Magnetization of GAP and $HoCu_2$ as a function of temperature for the magnetic field of 1 T.

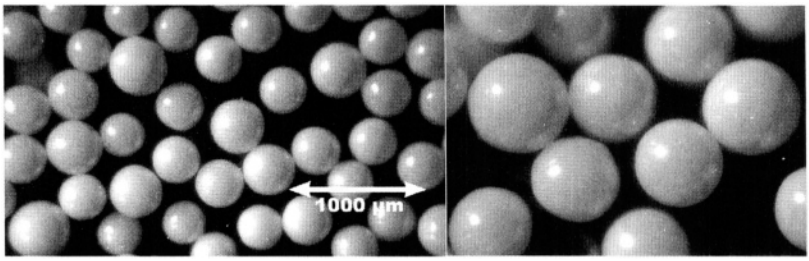


Figure 8. Photographs of fabricated GOS particles with smooth surface.

previously reported.[1] Starting raw material is very important especially in the GOS because the composition of sulfur is easily changeable. First we prepare the Gd_2O_3 ultra-fine powders with diameter of ~0.3 μm which is produced by a wet chemical processing. Then, the powders are reacted chemically with hydrogen sulfide gas to produce Gd_2O_2S powders directly. This simple method is useful to obtain the uniform fine powders of GOS. The granulator is used to make the spherical particles between 0.2 mm and 0.5 mm in diameter. Then, the particles are calcined at a temperature of 1500 °C in Ar atmosphere. After choosing the proper size particles with screen meshes, the chosen particles are polished to remove the thin and weak surface layer.

Precise control technique has been improved to produce more spherical particles with the smooth surface. Currently the distribution range of the particle size can be reduced within 0.1 mm and the larger particle size is obtainable; typical average particle size is 0.4 mm ± 0.05 mm. Figure 8 shows the photographs of the fabricated GOS particles with the smooth surface. The relative density to that of the single crystal is 99 % and Vickers hardness is ~900 which is a high enough value used as the regenerator material.

EXPERIMENTAL RESULTS ON COOLING TEST

GM cycle and pulse tube cryocoolers were used to test the cooling performance of the GOS regenerator material. For the GM cryocooler, two kinds of commercial models, Sumitomo SRDK-408D (1W@4.2K) and Sumitomo SRDK-101D (0.1W@4.2K) were chosen. The pulse tube cryocooler was a prototype model with a cooling power of 0.5 W at 4.2 K. The cooling tests were done by replacing the $HoCu_2$ with the GOS at the 2nd regenerator. Note that the initial condition of the 2nd regenerator consisted of $HoCu_2$ and Pb and we did not change the volume of

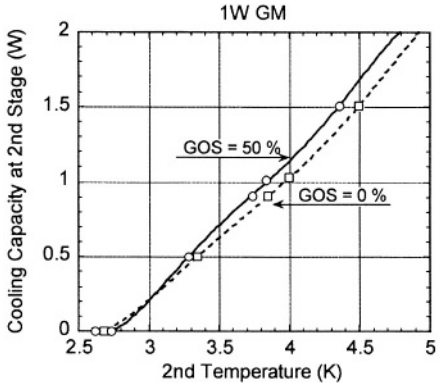


Figure 9. The experimental result of temperature dependence of cooling capacity at 2nd regenerator for the 1W GM cryocooler.

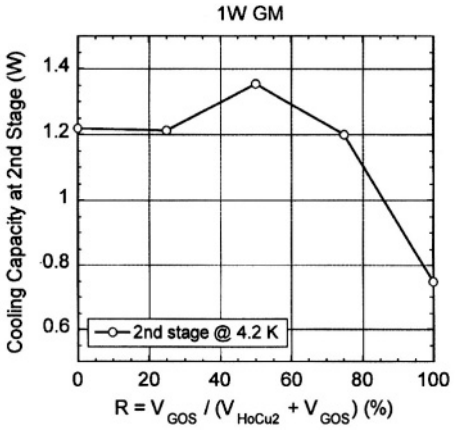


Figure 10. Volumetric ratio R dependence of the cooling capacity of the 2nd regenerator at 4.2 K for the 1W GM cryocooler.

the Pb. In order to make clear the replacing effect of the GOS, we introduce the volumetric ratio R, which is defined by the ratio of the volume of GOS to the volume of $(HoCu_2+GOS)$ at the lower temperature portion of the 2nd regenerator. Details on the experimental procedure and the measured results will be shown in our other papers.[7,8]

Figure 9 shows the experimental result on the temperature dependence of the cooling capacity at the 2nd regenerator for the 1 W GM cryocooler. At R = 50 %, the cooling capacity increases ~10 % at 4.2 K. It is noticeable that the cooling capacity keeps increasing even above 4.5 K, where it decreases largely in the case of the GAP. Figure 10 shows the R dependence of the cooling capacity of the 2nd regenerator at 4.2 K for the 1 W GM cryocooler. It is clear that an optimum value of R can be found around 50 % to provide the maximum cooling capacity at 4.2 K. This behavior is different from the case of the GAP in which the optimum R is ~25 %. For R > 60 %, the cooling capacity decreases largely. This is because the heat capacity the GOS steeply decreases above 5.5 K and the compensation for lack of the heat capacity is required by using some amounts of the $HoCu_2$.

Figure 11 shows the R dependence of the cooling capacity at 4.2 K for the 0.5 W pulse tube

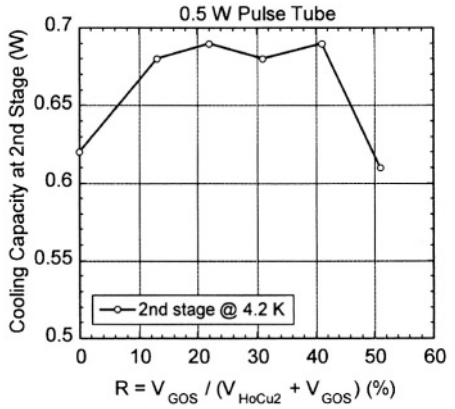


Figure 11. Volumetric ratio R dependence of the cooling capacity of the 2nd regenerator at 4.2 K for the 0.5W pulse tube cryocooler.

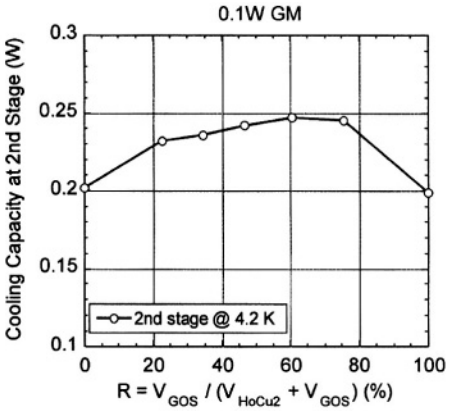


Figure 12. Volumetric ratio R dependence of the cooling capacity of the 2nd regenerator at 4.2 K for the 0.1W GM cryocooler.

cryocooler. The cooling capacity curve shows a local minimum between R = 20 and R = 40, but we guess that it may be caused by the setting error of the experiment. For R > 50 %, the cooling capacity decreases largely less than that of R = 0 and it is clear that an optimum R will be found for R < 50 %. This result is different from the case of the 1 W GM cryocooler; fewer amounts of the GOS will be enough to improve the cooling capacity of the pulse tube cryocooler.

For the 0.1 W GM cryocooler, the cooling capacity as a function of R has a broad peak as shown in Fig. 12. The peak value can be found around R = 65 % where the increasing rate on the cooling capacity of 25 % is obtainable. It is somewhat surprising that the cooling capacity does not decrease much at R = 100 % without any $HoCu_2$. Therefore, the GM cryocoolers will require more amounts of the GOS than the pulse tube cryocooler to increase the cooling performance.

CONCLUSIONS

A new generation magnetic regenerator material GOS has been developed for 4 K cryocoolers. The fabrication techniques are progressing to produce more uniform sphere particles with the sufficient hardness as the regenerator material, and additionally, to realize less cost compared with the conventional metallic magnetic regenerator materials. Long term tests with the commercial cryocoolers are proceeding. No broken particles have been found and the temperature stability of ± 45 mK at the 2nd stage regenerator with the heat load of 1 W has been measured for the continuous operation at the elapsed time of one month.

Variety of other ceramic magnetic regenerator materials are under development to cover the wider temperature ranges. Also new innovative approaches are being studied to make the regenerator configuration much simpler by using the flexible fabrication of ceramics.

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